

Revelation: An AI Agent-Powered Framework for Holistic Interpretation of IoT Network Traffic

Daniel Adu Worae and Spyridon Mastorakis

Abstract—Modern Internet of Things (IoT) networks generate heterogeneous, high-volume traffic that reflects both routine operation and evolving threats. Translating this telemetry into actionable insight requires interpretation across protocols, behaviors, and contextual signals, not detection alone. This paper presents *Revelation*, a framework that transforms raw packet captures into structured, semantically enriched artifacts to support operator-facing understanding and interactive analysis. *Revelation* employs a modular workflow for protocol and behavior reconstruction, packet- and flow-level structuring, transformer-based anomaly identification, selective threat-intelligence enrichment for public endpoints, and session-aware indexing for retrieval-augmented question answering. At query time, an agent retrieves relevant evidence from the indexed artifacts and composes grounded responses by coordinating the appropriate analysis steps. The framework captures both benign and adversarial activity and produces operator-readable summaries grounded in traffic context. We evaluate *Revelation* on four representative traffic scenarios (benign activity, uploading or exfiltration-like behavior, backdoor activity, and HTTP-based DDoS) using an artifact-grounded bank of 160 question-answer pairs. Six open-source models are tested under identical prompts in two retrieval settings: a dense-only baseline and a hybrid lexical-semantic configuration with keyword fallback and reranking. Across BLEU, ROUGE-1/2/L, METEOR, and BERTScore, hybrid retrieval consistently improves answer quality while reducing verbosity across all scenarios and models. We further profile execution time, CPU utilization, GPU memory usage, system memory usage, and token statistics to characterize efficiency and overhead. Overall, *Revelation* enables accurate, evidence-grounded interpretation of IoT packet captures.

Index Terms—IoT, network traffic analysis, anomaly detection, LLMs, AI agent, RAG, traffic interpretation

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I. INTRODUCTION

The Internet of Things (IoT) continues to expand at a massive scale, with billions of interconnected devices generating continuous streams of network telemetry [1]. These devices span smart home hubs, industrial sensors, medical monitors, and autonomous vehicles, each with distinct communication protocols, lifecycles, and behavioral patterns. While this heterogeneity enables new services and automation, it also complicates operational visibility. Operators must routinely determine whether observed behavior reflects expected functionality, misconfiguration, failure, or malicious activity,

and connect low-level telemetry to actionable operational explanations.

Most existing approaches to IoT traffic analysis remain primarily detection- and classification-oriented. They apply statistical techniques, signatures, or supervised learning to label traffic segments, for example, by device type, protocol, or benign versus malicious behavior. Although effective for coarse characterization and for surfacing certain threats, these methods fall short in two fundamental ways. First, they provide limited transparency and context, leaving operators with isolated labels or alerts that lack supporting evidence and actionable provenance. Second, they are functionally narrow, optimizing for detection and coarse labeling rather than an integrated, cross-layer understanding of network activity. As a result, practical questions that arise during troubleshooting and triage, such as *which external endpoints were contacted during anomalous behavior or what communication patterns preceded a suspected incident*, are difficult to answer reliably within conventional workflows.

In parallel, recent advances in large language models and retrieval-augmented generation (RAG) have demonstrated strong performance on structured reasoning and question answering tasks [2]–[8]. However, network telemetry is not naturally aligned with the assumptions of language models. Packet captures and derived logs are large, irregular, and semantically sparse without protocol-aware structuring and contextualization. Directly prompting a language model with raw traces or uncurated logs often leads to brittle behavior, incomplete reasoning, or unsupported explanations. In operational settings, these failure modes are especially problematic because inaccurate interpretations can mislead response actions and erode user trust.

This paper presents *Revelation*, a workflow for holistic interpretation of IoT network traffic from packet captures. *Revelation* converts raw captures into a structured, semantically enriched corpus that links packet-level observations, flow-level behaviors, protocol semantics, and endpoint context. The resulting artifacts are indexed for retrieval so that responses can be grounded in traceable evidence rather than generated from free-form speculation. Interactive analysis is guided by an agent that operates over the indexed corpus and coordinates specialized analysis functions at question time to locate relevant evidence and assemble a coherent, capture-grounded answer. When a question is outside the scope of capture-derived evidence but remains within the IoT domain, the agent consults authoritative external references and returns a sourced response that is clearly distinguished from trace-derived findings. This separation is intended to

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¹Code: <https://github.com/WadElla/Revelation>

preserve evidentiary clarity and to reduce the risk of conflating external knowledge with local observations.

Revelation supports a full-spectrum understanding of network activity. It summarizes routine behavior as well as adversarial patterns, connects observations across packet-, flow-, and protocol-level views, and produces operator-readable interpretations that emphasize what occurred, why it matters, and what actions may be appropriate. These capabilities support practical workflows such as incident triage, configuration validation, and compliance-oriented reporting.

To frame the technical scope and foundational impact of *Revelation*, we pose the following research questions:

- **RQ1:** What is required to move beyond anomaly detection toward full-spectrum interpretation of IoT traffic, covering benign and malicious behavior across diverse protocols and device roles?
- **RQ2:** To what extent does an agent that processes and retrieves from structured packet-capture representations outperform a language model that lacks traffic-processing and evidence-retrieval capabilities for IoT traffic interpretation?
- **RQ3:** Which retrieval strategy yields higher answer quality over the same corpus and prompts: dense retrieval alone or a hybrid lexical-semantic approach with fallback mechanisms and reranking?
- **RQ4:** Can *Revelation* provide accurate, context-sensitive answers while remaining efficient and responsive under realistic workloads, as measured by execution time, CPU and memory usage, GPU memory usage, token counts, and response sizes?

In addressing these questions, we contribute a novel architecture and set of capabilities that redefine how IoT network telemetry can be interpreted and operationalized:

- We present *Revelation*, a framework that converts packet captures into a structured, queryable corpus to enable evidence-grounded interpretation of IoT traffic. The workflow produces complementary artifacts spanning packet-level structure, flow-level summaries with endpoint context, anomaly interpretation from a fine-tuned transformer classifier, and selective enrichment for public endpoints, supporting analysis of both benign and adversarial behaviors.
- We develop an agent-driven question-answering mechanism that coordinates capture-grounded retrieval and analysis at query time and uses a distinct source-lookup pathway for IoT questions when local evidence is weak or out of scope.
- We evaluate *Revelation* on a 160-question benchmark spanning four packet-capture scenarios and six open models under matched prompts and a shared indexed corpus. A hybrid retrieval configuration consistently outperforms a dense-only baseline across BLEU [9], ROUGE-1/2/L [10], METEOR [11], and BERTScore [12].
- We profile deployability through an operational study of execution time, CPU utilization, GPU memory, system memory, token counts, and response sizes in both retrieval settings, demonstrating efficiency suitable for local

deployment with open models.

- We release the full implementation, source code, and data as open source to support transparency and reproducibility: <https://github.com/WadElla/Revelation>

The rest of our paper is organized as follows: Section II describes the system design of *Revelation*. Section III presents experimental results and an analysis of *Revelation*. Section IV highlights the Security and Privacy Analysis of *Revelation*. In Section V, we present a literature review of *Revelation*. Finally, we conclude in Section VI.

II. SYSTEM DESIGN

Revelation is a workflow for holistic interpretation of IoT network traffic. It transforms packet captures into a structured, indexed collection of traffic-derived artifacts, including protocol logs, packet-layer records, flow summaries, a transformer-based anomaly report, and selective threat intelligence annotations, and prepares them for retrieval. Interactive analysis is mediated at question time by an agent that uses tools to search this indexed collection and, when a query is IoT-focused but outside the capture, consults authoritative sources. This section outlines the core components and explains how the workflow supports clear, extensible, and operationally relevant analysis.

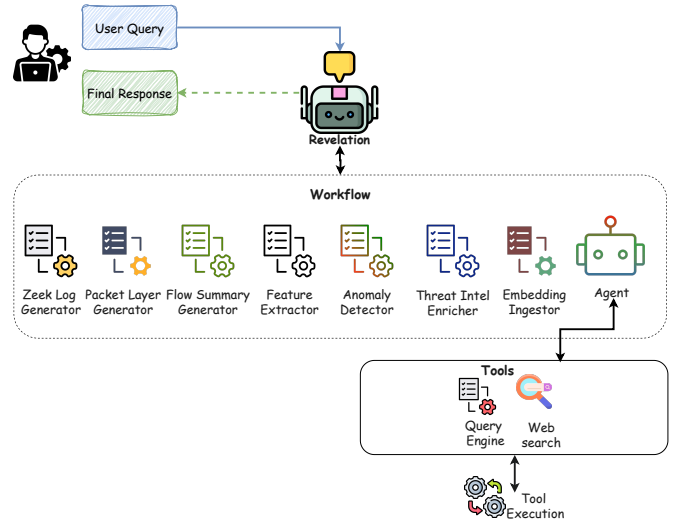


Fig. 1: System Design of *Revelation*

A. PCAP Ingestion and Protocol Log Generation

Revelation begins by ingesting a user-provided packet capture and extracting protocol-level activity into structured, per-protocol JSON logs. In our implementation, we use Zeek to generate these logs in a standard JSON log schema [13]. This stage reconstructs protocol semantics from packet streams and records them as timestamped events suitable for indexing and downstream analysis. The resulting logs summarize protocol-layer behavior commonly observed in IoT environments, including connection establishment and application transactions for services such as name resolution, web communication, message-oriented telemetry, industrial control exchanges, and encrypted transport. Each event preserves original timestamps

and salient protocol fields needed to characterize requests and responses with fidelity.

These protocol logs are persisted and indexed alongside packet-layer records, flow summaries, and the anomaly interpretation report, forming a unified retrieval surface for operator queries. By exposing protocol semantics as searchable evidence, *Revelation* supports explanations grounded in concrete network events and traceable to the underlying capture.

B. Packet-Layer Representation

Protocol-level records provide compact behavioral summaries, but many operator questions require packet-granular evidence, such as validating handshake progression, confirming request–response sequencing, and inspecting transport control behavior when higher-level reconstruction is incomplete or ambiguous. *Revelation* therefore builds a packet-layer view that preserves low-level structure with minimal abstraction.

Each packet is exported as a structured JSON record reflecting the layered header hierarchy from link through transport, and it preserves application-layer header fields whenever they are observable. Records preserve original timestamps and capture order, size information, endpoint identifiers, and transport state signals, including control flags and sequence-related metadata. When application headers are available, the representation retains only the minimal fields needed to convey intent, without embedding payload content.

To maintain interpretability and limit token overhead, *Revelation* applies targeted normalization. High-entropy binary fields and opaque payload bytes are removed, including raw segment contents and cryptographic randomness, while semantically meaningful header fields are preserved. The resulting packet records remain faithful to protocol structure and timing without introducing extraneous binary artifacts.

Finally, packets are grouped into retrieval-ready segments using context-aware segmentation aligned with conversational boundaries and protocol events, including handshake phases, request–response groupings, and time-gap delimiters. This packet-layer representation provides verifiable evidence for fine-grained queries and complements higher-level summaries by enabling precise cross-checking against observable packet behavior.

C. Flow Summary Generator

The flow summary generator reconstructs bidirectional sessions from the packet capture for both TCP and UDP. A flow is defined as the set of packets exchanged between two endpoints that share the same transport protocol and are associated with a consistent endpoint pair, identified by the source and destination addresses and ports. Packets in both directions are grouped into a single session-level unit. This representation elevates analysis from individual packets to conversational behavior, which more closely matches how operators interpret intent, stability, and anomalies.

For each flow, the generator records endpoint addresses and ports, transport protocol, packet and byte counts, start and end times, and duration. When application-layer headers are observable, the summary incorporates lightweight indicators that

provide functional context, such as request types, identifiers, and selected protocol fields. This preserves interpretability without requiring payload inclusion.

Connection dynamics are captured by decoding transport control behavior. For TCP flows, flag sequences are summarized to reflect handshake completion, resets, and early termination, providing concise signals of session health that complement application cues.

When flows involve public endpoints, the generator enriches the summary with reputation information obtained from AbuseIPDB [14]. The inline annotation includes the abuse-confidence score to distinguish benign from potentially suspicious activity. When link-layer information is available, MAC addresses are mapped to vendor identifiers via Organizationally Unique Identifier (OUI) parsing to support device attribution.

The output is a structured, operator-readable narrative in which each flow is represented as a standalone block. These blocks are segmented into retrieval-ready units and indexed as part of the overall evidence corpus. By combining session-level statistics, protocol cues, connection-state signals, and targeted enrichment, flow summaries support fine-grained retrieval and enable tasks such as forensic investigation, incident triage, and validation of expected device behavior.

D. Feature Extractor

This stage derives model-ready features directly from the packet capture using protocol-aware inspection. It computes a fixed set of statistical, temporal, and protocol-specific attributes relevant to IoT environments, drawing from TCP/IP and application protocols frequently observed in practice, including DNS, MQTT, and Modbus.

The capture is transformed into a tabular representation in which each row corresponds to a packet, and each column encodes a well-defined feature. The feature set includes traffic descriptors such as packet and byte counts and durations, transport indicators such as control-flag behavior, and application-layer fields when observable, for example request types and protocol operation identifiers.

For transformer-based training and inference, each row is then serialized into a single textual sequence of feature–value pairs. Every column receives a descriptive label, followed by a delimiter and the observed value, and pairs are concatenated in a canonical order to form one compact input string. For example, a destination port may appear as `tcp.dstport:442`, with subsequent pairs appended using consistent separators. This textualization preserves the semantics of the tabular representation, enables straightforward handling of mixed data types, and provides a uniform interface for language models to process packet- and flow-level context.

E. Semantic Anomaly Detection and Reporting

This stage applies a fine-tuned transformer classifier to the textual feature sequences produced in **Section D** to categorize traffic as benign or as one of multiple attack classes spanning reconnaissance, injection, denial of service, and persistence.

Beyond per-sample predictions, the module generates an interpretation report that summarizes traffic composition, highlights dominant threats, and explains observed behaviors in operator-readable prose. For each detected attack type, the report emphasizes relevant protocol cues, provides concise guidance, and aggregates contextual metadata that supports follow-up investigation, including communicating endpoint pairs and selected application indicators when available.

Model training and configuration. The anomaly detector is fine-tuned using benchmark IoT traffic datasets. We fine-tune on Edge-IIoTset [15] and additionally fine-tune and evaluate on TON_IoT [16] to assess adaptability across heterogeneous environments. During preprocessing, high-cardinality identifiers and dataset-specific fields are removed to reduce overfitting and improve generalization. Each example is serialized into a canonical sequence of labeled feature–value pairs, tokenized using the BERT tokenizer, and passed to a sequence-classification head configured to the corresponding label set. Fine-tuning follows supervised learning and is evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score, with class-wise performance reported in a detailed classification report.

Outputs. The module produces two artifacts: (i) a structured CSV containing per-sample predictions and (ii) a narrative interpretation report intended for operator use. The report provides class-level summaries supported by protocol evidence and compiles endpoint and application metadata for each detected category. These outputs support situational awareness and provide structured inputs for subsequent enrichment and retrieval.

F. Threat Intelligence Enrichment

The threat-intelligence enrichment stage identifies public IP addresses referenced in the anomaly interpretation report and augments them with external context. For each public IP, *Revelation* queries two complementary sources: a malware and reputation aggregator (VirusTotal [17]) and an Internet exposure snapshot service (Shodan InternetDB [18]). Retrieved signals include aggregated detection counts, evidence of recent malicious activity, observed open TCP/UDP ports and exposed services, descriptive tags, and known CVE identifiers when available. Each public IP is mapped to its associated attack category, and a concise, structured summary is inserted into the corresponding metadata block of the report.

This contextualization connects capture-derived findings to externally observable exposure and reputation signals. For example, an address implicated in scanning activity may present externally exposed administrative services, while an address associated with traffic predicted as backdoor may carry multiple reputation flags. Enrichment is applied only when public IPs are present to avoid unnecessary external queries and to preserve provenance clarity.

The enriched report is then stored as part of the evidence corpus, enabling downstream retrieval to surface both the local behavioral evidence and the associated external context in a single, clearly attributed response.

G. Vectorization and Semantic Ingestion

To enable evidence-grounded reasoning across heterogeneous IoT traffic representations, *Revelation* ingests traffic-derived artifacts into a unified semantic index. The ingestion layer converts protocol-level records, packet-layer structures, flow summaries, and anomaly interpretation reports into retrieval-ready chunks, embeds them into a shared vector space, and stores them with compact provenance metadata. This design supports cross-layer querying, allowing the system to retrieve high-level explanations and, when needed, return to fine-grained packet or protocol evidence.

1) *Multi-View Artifact Integration:* *Revelation* ingests four complementary modalities that reflect distinct abstraction levels of the same capture: (i) protocol-level records that summarize reconstructed events and transactions, (ii) packet-layer structured representations that preserve header-level details and timing, (iii) flow summaries that capture session-level statistics, control behavior, and endpoint context, and (iv) anomaly interpretation reports that consolidate predicted classes with operator-readable explanations and enriched metadata. Treating these artifacts as a single evidence corpus enables retrieval to traverse between behavioral summaries and the underlying observations that justify them.

2) *Modality-Aware Chunking and Metadata:* Because each modality has a different structure and granularity, *Revelation* applies modality-aware segmentation to produce coherent retrieval units. Protocol-level records are grouped by session identifiers so that all events belonging to the same connection or transaction are retrieved together. Flow summaries are treated as atomic units, with one block per reconstructed flow. Interpretation reports are segmented by functional structure into retrieval units that isolate summary findings, explanatory narratives for detected activity, and the associated metadata blocks. Packet-layer structures are segmented using semantic grouping that clusters packets describing the same interaction into coherent, retrieval-ready units. This grouping follows conversational and event boundaries, such as handshake phases, request–response sequences, and short temporal neighborhoods, so that each segment reflects a meaningful slice of packet behavior rather than an arbitrary size-based split.

Each chunk is annotated with a compact metadata schema encapsulating the source modality, abstraction level, and a stable internal identifier for deduplication and traceability. The schema preserves contextual integrity across modalities while preventing disclosure of network-identifiable attributes.

3) *Embedding, Storage, and Session Control:* Following modality-specific segmentation, each chunk is embedded using a locally hosted encoder to map heterogeneous artifacts into a shared semantic space. The embeddings, chunk text, and essential metadata are stored in a session-isolated vector index. Each ingestion run produces a uniquely identified session that serves as a consistent snapshot of the evidence corpus for a given capture and configuration. Interactive queries resolve the active corpus via a latest-session pointer, while prior sessions remain available for comparison and reproducibility. To bound storage and indexing overhead, *Revelation* retains only the three most recent sessions and removes older sessions according to a least-recently-used policy.

4) *Deduplication and Session Management*: The framework avoids redundant computation through file-level content hashing at two points in the pipeline. First, a capture-level hash is used as a skip guard for upstream processing. If an identical capture has already been processed in the most recent session, previously produced artifacts are reused. Second, before embedding, the ingestion layer computes file-level hashes for each artifact source and compares them against the hashes from the most recent three sessions. If the artifact set matches an existing session, ingestion is skipped, and the active-session pointer is updated to reference the corresponding stored index. A lightweight session index maps stable dataset identifiers (e.g., PCAP content hashes) to their associated vector stores and creation times. Interactive queries resolve the active session through this index, ensuring that retrieval operates on the latest consistent context while still allowing explicit selection of historical sessions for side-by-side analysis.

H. Contextual Retrieval and Query Engine

The query engine maps natural-language questions to the indexed traffic artifacts and retrieves evidence suitable for interactive analysis, including incident triage and contextual troubleshooting. Given a query and an active session, the engine returns a ranked set of evidence chunks drawn across abstraction levels, including protocol-level records, flow summaries, packet-layer structures, and anomaly interpretation reports (Figures 2 and 3). Retrieval is session-aware, targeting the most recently ingested capture by default, and the resulting context is passed to the agent for answer synthesis.

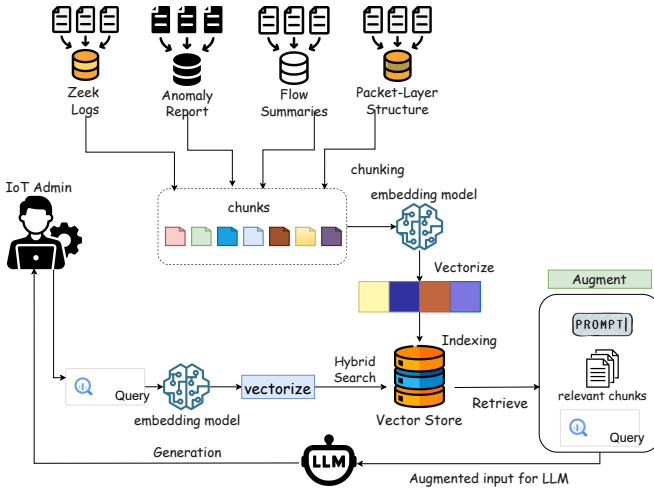


Fig. 2: Retrieval-Augmented Generation Pipeline

Multi-Stage Retrieval Pipeline

1) *Query Embedding*: The user query is embedded using the same encoder applied during ingestion to place queries and evidence into a compatible representation space. This alignment guarantees that abstract queries (e.g., “How is MITM behavior reflected in the traffic metadata?”) can be mapped meaningfully to technical evidence across packet-layer records, flow summaries, protocol logs, and interpretation reports.

2) *Hybrid Semantic Retrieval*: To maximize retrieval fidelity, the engine performs a hybrid search that integrates three complementary signals, as shown in Figure 3:

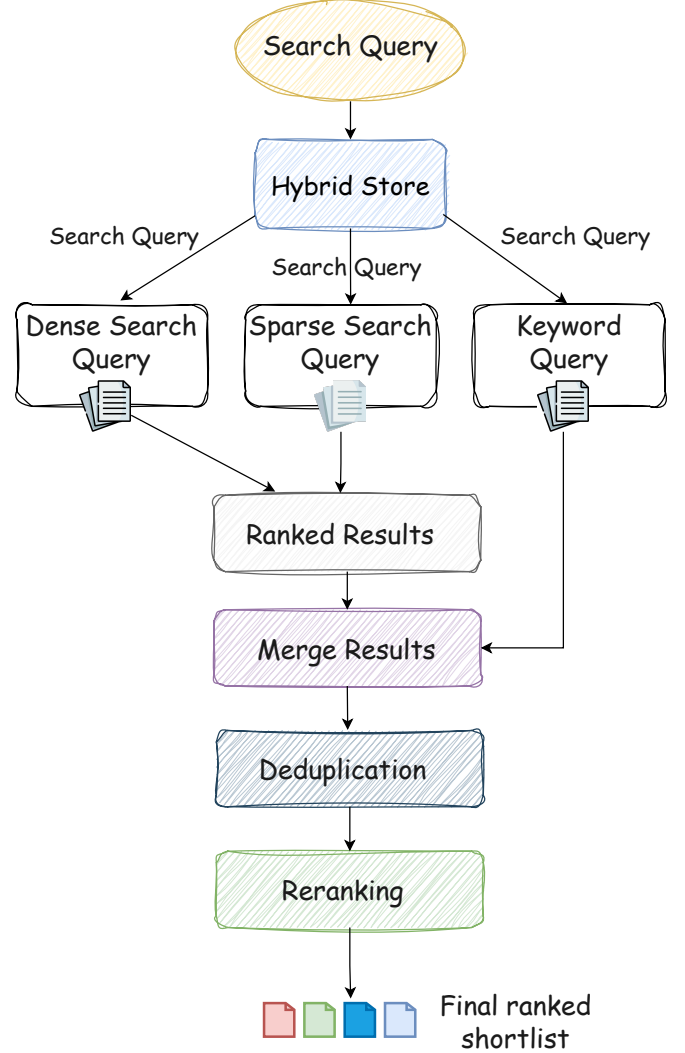


Fig. 3: Hybrid Semantic Retrieval and Reranking Workflow

- **Dense Semantic Search**: Using cosine similarity, the query vector is compared to stored chunk embeddings. This captures latent conceptual relationships, enabling the system to surface results even when terminology differs (e.g., “unauthorized access” vs. “intrusion”).
- **Sparse Lexical Search (BM25)**: Simultaneously, a BM25 index built over the same document corpus retrieves chunks based on token overlap and term frequency. This enhances precision, particularly for structured logs or keyword-driven forensic queries.
- **Literal Keyword Matching (Fallback)**: As a robustness mechanism, the engine also applies direct keyword matching on chunk contents. This ensures coverage in cases where both dense and sparse models fail due to edge-case phrasing or rare entity references.

Dense and sparse scores are interpolated using a tunable weighting function, and the resulting top-ranked hybrid candidates are merged with fallback matches from the keyword

stage to form a comprehensive evidence set. This set is passed downstream for ranking.

3) *Evidence Consolidation and Reranking*: Candidate chunks from the hybrid retrieval stage are deduplicated using content-based hashes to eliminate redundancy. A cross-encoder reranker then performs pairwise relevance scoring between the query and each chunk. This step models deep inter-sentence interactions, producing a final ranked shortlist of the most contextually relevant evidence for the query.

4) *Session Awareness and Temporal Consistency*: The Query Engine maintains alignment with the most recent ingestion session to ensure that retrieval reflects the latest observed traffic state. Session boundaries are tracked automatically, and only a bounded number of recent sessions are retained to control storage and indexing overhead while still supporting limited historical comparison.

I. Agent for Question Answering

The agent constitutes the reasoning layer that transforms an operator’s query, together with the ranked context returned by retrieval, into a precise, well-sourced answer. Operating over the active session, it ingests the top-k evidence selected by the query engine and treats this material as its working state. Rather than issuing a one-shot response, the agent executes a compact perceive–reason–act cycle that incrementally resolves the query while remaining aligned with available context.

Planning and gap resolution. The agent employs a language model to infer intent, decompose composite requests, and identify residual gaps after initial retrieval. When the evidence set is sufficient, planning reduces to consolidation and articulation of the key points. When coverage is incomplete, the agent specifies narrowly targeted follow-ups—refining retrieval with more discriminative terms or, for IoT-domain details that fall outside the processed capture, consulting an authoritative external source.

Bounded actions. The agent employs a bounded toolset that aligns with its operational scope. Its default action is to invoke the retrieval-and-answer tool, which merges top-ranked evidence from the indexed corpus and drafts an answer based on the relevant chunks. When the query is IoT-focused but clearly beyond the scope of the capture (e.g., asking about protocol standards, mitigation strategies, or CVE details), or when retrieved evidence is insufficient, the agent invokes a complementary web-lookup tool. This targeted query ensures that external, authoritative sources are consulted in a controlled manner, yielding citable snippets that complement locally indexed data without diluting precision.

Faithfulness and attribution checks. Before finalization, the agent conducts lightweight faithfulness checks. These verify that each claim is supported by the cited passages, ensure consistency of protocol terminology and identifiers, and screen for unwarranted generalization. If support is lacking or ambiguity persists, the agent revises its plan and repeats the minimal necessary retrieval or lookup. The process terminates when the response is adequately supported, appropriately sourced when external knowledge is used, and directly responsive to the operator’s intent.

Agent instructions. The agent is initialized with concise system-level instructions that (i) treat the top-k retrieved context as primary evidence and avoid unsupported speculation; (ii) answer capture-grounded queries using only the provided context, stating explicitly when information is unavailable; (iii) allow pretrained knowledge for clearly general IoT questions when the claim is stable and time-insensitive (e.g., protocol semantics, standard ports); (iv) require a focused web lookup and brief citation for time-varying or source-dependent facts; and (v) enforce plain-text, technically precise, and concise responses, finalized only after a self-check confirms support. At synthesis time, the agent governs orchestration using its internal instructions. When it invokes the retrieval-and-answer tool, it receives a package that has the system prompt, the user query, and the top-k context to frame the Large language model’s behavior.

This design provides disciplined autonomy at query time. By reasoning over session-scoped context, selecting minimal actions to close evidence gaps, and enforcing faithfulness and attribution, the agent produces concise and defensible answers aligned with the indexed traffic artifacts.

1) *Conversation Session Management: Revelation* maintains two distinct forms of state: dialogue state, which captures the conversational context across turns, and retrieval state, which identifies the active evidence corpus produced from a specific ingestion session. The system supports two modes. In capture-bound mode, the dialogue session is deterministically bound to the active retrieval session for a given packet capture, preserving continuity for follow-up questions while preventing cross-capture context leakage. In free mode, the analyst initiates an independent dialogue thread (user-named or auto-generated) whose memory is decoupled from any capture, enabling exploratory or comparative analysis without inheriting capture-specific narrative context. In both modes, dialogue histories are persisted, reviewable in the interface, and independently purgeable; when retrieval is invoked, it is resolved against the currently selected capture session to preserve provenance and reproducibility.

III. EXPERIMENTAL RESULTS AND ANALYSIS

This section evaluates *Revelation* along two dimensions. First, we assess artifact-grounded question answering over packet-capture representations by comparing a dense-only retriever with a hybrid lexical–semantic configuration that includes keyword fallback and cross-encoder reranking under matched prompts and a shared indexed corpus. Second, we evaluate semantic anomaly detection using a fine-tuned transformer classifier, reporting accuracy, precision, recall, F1-score, and class-wise performance. Together, these experiments quantify how retrieval design influences answer quality and efficiency and characterize the reliability of the detection backbone that informs downstream interpretation.

A. PCAP-Grounded Question Answering

1) Experimental Setup

We analyze four representative PCAPs covering benign and adversarial scenarios: Normal traffic, Uploading attack, Backdoor attack, and DDoS HTTP flood. Each capture is processed

TABLE I: Performance Evaluation of *Revelation*'s Retrieval Variants across IoT PCAPs

PCAP	Metric	Gemma2 (Dense)	Gemma2 (Hybrid)	Llama 3.2 (Dense)	Llama 3.2 (Hybrid)	Mistral (Dense)	Mistral (Hybrid)	LLaVA (Dense)	LLaVA (Hybrid)	Qwen3 (Dense)	Qwen3 (Hybrid)	Phi-4-mini (Dense)	Phi-4-mini (Hybrid)
Normal traffic	BERTScore (p/r/f)	79.83 / 84.97 / 82.29	86.52 / 85.34 / 85.87	80.10 / 85.61 / 82.73	86.28 / 88.31 / 87.24	81.60 / 85.99 / 83.71	87.33 / 88.60 / 87.91	84.14 / 87.39 / 85.69	88.13 / 89.63 / 88.82	77.00 / 84.94 / 80.76	78.40 / 85.91 / 81.96	69.56 / 74.39 / 71.85	84.10 / 87.76 / 85.85
	ROUGE (r1/r2/rL)	14.68 / 5.45 / 11.37	36.34 / 23.22 / 34.43	14.86 / 6.03 / 12.24	41.14 / 24.44 / 35.94	17.26 / 7.83 / 14.41	42.12 / 27.11 / 38.17	28.97 / 14.63 / 25.44	49.97 / 33.21 / 45.98	3.39 / 1.56 / 3.04	10.20 / 6.36 / 9.57	3.40 / 0.95 / 2.95	30.92 / 17.00 / 26.72
	BLEU	0.93	1.37	1.22	9.06	1.64	13.24	4.04	15.55	0.21	2.15	0.10	5.69
	METEOR	16.23	21.51	17.03	33.61	18.81	34.23	25.67	38.83	6.24	19.77	1.48	30.23
Uploading attack	BERTScore (p/r/f)	79.78 / 85.39 / 82.46	86.36 / 85.65 / 85.96	79.85 / 85.67 / 82.63	85.76 / 88.13 / 86.87	81.75 / 86.58 / 84.07	86.44 / 89.46 / 87.88	84.34 / 87.55 / 85.89	87.35 / 88.84 / 88.04	76.60 / 84.98 / 80.56	78.52 / 86.11 / 82.12	71.47 / 78.11 / 74.60	83.82 / 86.31 / 84.97
	ROUGE (r1/r2/rL)	14.58 / 5.74 / 12.19	34.35 / 20.08 / 32.46	15.87 / 7.56 / 13.22	38.62 / 24.34 / 34.86	17.97 / 8.04 / 14.48	40.19 / 27.05 / 36.84	26.84 / 14.65 / 23.77	44.93 / 29.60 / 41.29	3.42 / 1.57 / 3.08	11.00 / 6.78 / 10.07	2.93 / 0.26 / 2.57	29.22 / 16.90 / 25.15
	BLEU	0.82	8.68	2.44	12.28	2.06	13.68	4.58	13.93	0.20	2.35	0.14	6.64
	METEOR	15.45	21.48	19.24	31.27	19.30	36.40	24.84	33.88	6.20	20.63	2.18	22.05
Backdoor attack	BERTScore (p/r/f)	80.06 / 84.26 / 82.07	86.10 / 84.30 / 85.16	81.16 / 86.01 / 83.47	86.20 / 87.50 / 86.80	82.34 / 85.54 / 83.88	86.90 / 88.31 / 87.57	84.91 / 87.47 / 86.13	89.16 / 88.94 / 89.01	77.36 / 84.64 / 80.82	78.78 / 85.95 / 82.19	71.91 / 77.62 / 74.61	83.07 / 85.68 / 84.28
	ROUGE (r1/r2/rL)	12.99 / 4.54 / 10.50	30.95 / 18.08 / 29.34	16.62 / 9.13 / 14.43	35.25 / 23.19 / 31.57	17.83 / 7.37 / 14.62	38.25 / 25.32 / 32.96	32.39 / 19.00 / 28.85	51.62 / 36.48 / 48.19	3.91 / 2.10 / 3.67	8.88 / 5.61 / 8.11	2.13 / 0.14 / 1.83	24.07 / 13.98 / 21.27
	BLEU	1.06	8.18	4.66	11.60	1.87	13.46	8.40	16.35	0.45	1.73	0.09	4.66
	METEOR	16.58	18.71	21.86	31.95	21.69	37.65	31.53	42.97	7.88	17.52	1.46	18.43
DDoS HTTP flood	BERTScore (p/r/f)	79.78 / 84.67 / 82.13	85.40 / 85.66 / 85.49	79.94 / 84.97 / 82.34	85.11 / 87.52 / 86.26	81.58 / 85.80 / 83.61	84.90 / 87.31 / 86.07	83.74 / 87.66 / 85.61	86.45 / 88.31 / 87.34	76.70 / 84.69 / 80.48	78.45 / 85.12 / 81.62	63.24 / 67.03 / 65.01	72.77 / 76.66 / 74.61
	ROUGE (r1/r2/rL)	13.26 / 4.75 / 11.10	31.27 / 17.68 / 29.57	13.67 / 5.61 / 11.25	34.00 / 19.24 / 30.10	17.53 / 7.77 / 14.66	32.04 / 17.45 / 27.81	31.72 / 16.49 / 28.60	44.92 / 25.72 / 39.98	3.78 / 1.75 / 3.51	10.38 / 5.59 / 9.25	1.75 / 0.00 / 1.32	20.37 / 12.58 / 18.51
	BLEU	0.80	8.18	1.18	9.54	1.92	8.10	4.17	12.60	0.27	1.75	0.08	4.69
	METEOR	15.10	20.47	18.40	30.13	20.76	29.85	29.70	39.32	7.48	19.41	0.34	18.50

Performance Metrics: p/r/f = Precision/Recall/F1-Score; ROUGE (r1/r2/rL) = ROUGE-1/ROUGE-2/ROUGE-L.

TABLE II: System Performance and Token Metrics Analysis

Metric	Gemma2 (Dense)	Gemma2 (Hybrid)	Llama 3.2 (Dense)	Llama 3.2 (Hybrid)	Mistral (Dense)	Mistral (Hybrid)	LLaVA (Dense)	LLaVA (Hybrid)	Qwen3 (Dense)	Qwen3 (Hybrid)	Phi-4-mini (Dense)	Phi-4-mini (Hybrid)
Execution Time (s)	3.1433	5.4415	1.5571	5.2264	1.9995	5.6687	1.5144	6.4263	18.4219	11.7037	3.3381	10.0193
Memory Usage (MB)	0.0575	7.7733	0.0497	7.7659	0.0590	7.8483	0.0606	7.8471	0.0544	7.8519	0.0645	7.7780
GPU Memory Used (MB)	0.0000	16.9607	0.0000	16.9607	0.0000	16.9607	0.0000	16.9607	0.0000	16.9607	0.0000	16.9607
CPU Utilization (%)	0.0334	2.4839	0.0638	2.6339	0.0512	2.3634	0.0620	2.4466	0.0267	1.4760	0.0535	2.2077
Avg. number of tokens	225.5867	35.4645	276.7226	76.9648	179.3652	79.0544	117.7630	46.7922	1631.4593	582.0061	401.2568	547.4785
Avg. Response size (bytes)	1185.8805	164.2447	1271.2372	361.9596	960.5727	401.3950	627.4910	222.8392	7089.2024	2593.3212	2317.1173	3608.0175

through the workflow into four evidence views: Protocol-Level Telemetry, Anomaly Interpretation Reports, Narrative Flow Summaries, and Packet-Layer Feature Views. For each PCAP, forty ground-truth question-answer pairs (ten per source) are compiled, yielding a benchmark of 160 references.

We evaluate six representative open-source models—Gemma2, LLaVA, Llama-3.2, Mistral, Qwen3, and Phi-4-mini. Two retrieval settings are compared under the same prompts: Dense-only, which ranks candidates based on embedding similarity, and Hybrid, which fuses dense retrieval with BM25 lexical search and keyword fallback before applying cross-encoder reranking.

2) Evaluation Metrics

In this question-answering and performance experiment, we employ text quality metrics to evaluate the generated answers and system performance metrics to measure efficiency.

Text Quality Metrics:

- **BERTScore (Precision/Recall/F1-Score)**: Evaluates semantic fidelity between generated answers and references using contextual embeddings. Precision measures the relevance of generated content, recall assesses coverage of reference meaning, and F1 balances the two.
- **ROUGE (1/2/L)**: Measures content overlap at unigram, bigram, and sequence levels. ROUGE-1 and ROUGE-2 track whether key entities and phrases are included, while ROUGE-L captures preservation of sentence structure and narrative order.
- **BLEU**: Focuses on n-gram precision with a brevity penalty, rewarding exact reproduction of technical phrases, protocol strings, and values where precision is critical.
- **METEOR**: Complements BLEU by balancing precision and recall while allowing for synonymy and stemming. It credits semantically correct paraphrases, ensuring robustness to linguistic variation.

System Performance Metrics:

- **Execution Time (s)**: The end-to-end latency measured from the moment the operator's query is received to the moment the final, consolidated answer is delivered.

- **Memory Usage (MB)**: Per-query change in resident set size (RSS) of the evaluation process over the query window, reported in MB and averaged across queries.
- **CPU Utilization (%)**: Fraction of logical cores used per query, computed as CPU time accrued during the query interval divided by wall-clock time and normalized by the number of logical cores.
- **GPU Memory (MB)**: Per-query change in device GPU memory usage over the query window, reported in MB and averaged across queries.
- **Average Tokens and Response Size (bytes)**: The mean number of output tokens generated per query is a direct measure of the language model's response verbosity, where reductions indicate tighter, more decisive responses. Token count and response length are computed using a fixed tokenizer to ensure consistent measurement across models and retrieval configurations.

3) Results Across PCAPs

As summarized in Table 1, Hybrid retrieval consistently and substantially outperforms Dense-only across all four PCAPs and six models. When averaged across all scenarios, BERTScore-F1 improves by 5.3%, while structure-sensitive metrics show even more dramatic relative gains: ROUGE-L improves by 146.6%, BLEU by 391.7%, and METEOR by 83.0%. These results confirm that Hybrid retrieval produces answers that are semantically faithful, structurally complete, and lexically precise, qualities that are essential for trustworthy interpretation.

Equally significant are the reductions in verbosity. Hybrid cuts the average number of tokens by 51.5% and reduces response size by 45.6%, yielding answers that are concise and more decisive. This efficiency has direct operational significance: when a user uploads a PCAP and queries the system, *Revelation* provides answers that are not only more accurate but also more concise, enabling administrators to act quickly without having to wade through redundant or verbose text.

The extent of improvements varies across traffic types, highlighting how *Revelation* adapts to different contexts. For

Normal traffic, Hybrid improves BERTScore-F1 by 6.3%, ROUGE-L by 164.3%, BLEU by 575.7%, and METEOR by 96.9%, while tokens fall by 71.6%. This shows that *Revelation* can summarize benign activity clearly and efficiently, helping operators confirm normal operations with minimal effort.

For Uploading attacks, Hybrid achieves BERTScore-F1 gains of 5.2% and significant relative improvements across ROUGE, BLEU, and METEOR. Token reduction is even sharper at 73.4%, ensuring that exfiltration cues are highlighted succinctly while retaining full accuracy.

For Backdoor scenarios, the improvements are more modest, with BERTScore-F1 increasing by 4.9%, ROUGE-L by 135%, BLEU by 265.1%, and METEOR by 66.9%. Token usage decreases by 24.5%, ensuring more precise answers that highlight persistence mechanisms with less redundancy, supporting analysts who need to track stealthy behaviors.

For DDoS HTTP floods, Hybrid retrieval improves BERTScore-F1 by 4.6%, ROUGE-L by 126.3%, BLEU by 544.6%, METEOR by 76%, and reduces the number of tokens by 22.4%. Although the relative improvements in terms of the number of tokens are smaller than in other cases, the system still provides more precise and less verbose answers. This ensures that operators facing large-scale volumetric attacks can extract essential insights without being overwhelmed by redundant descriptions.

Taken together, these results demonstrate that *Revelation*'s retrieval pipeline is not only technically superior but also operationally feasible. Across benign baselines, stealthy persistence, data exfiltration, and volumetric floods, the system consistently generates answers that are more accurate, concise, and aligned with the holistic interpretation needs. This means that when an administrator uploads a PCAP, *Revelation* provides outputs that are both trustworthy in content and practical in form, ensuring that the system can be deployed as a usable tool for interactive IoT traffic analysis.

4) System Performance Analysis

System behavior, summarized in Table 2, shows that the computational overhead of Hybrid retrieval is modest and predictable. Execution time increases by 52.3%, reflecting the additional retrieval and reranking steps. Memory usage rises by +7.75 MB, GPU memory by +16.96 MB, and CPU utilization by +2.22%. These absolute increases remain small in the context of administrative systems, confirming that *Revelation* can scale without resource bottlenecks.

The largest efficiency gains are observed in the generation footprint. Tokens are reduced by 51.5% and response bytes by 45.6%, significantly lowering verbosity. This not only offsets retrieval overhead but also improves throughput in multi-query workloads, ensuring that users uploading large PCAPs can obtain timely and actionable answers. In some cases, the shorter outputs even compensate for retrieval overhead, producing neutral or faster end-to-end response times.

Hybrid retrieval significantly enhances the accuracy, conciseness, and interpretability of answers while incurring only minor and bounded resource overhead. This makes *Revelation* well-suited for interactive, PCAP-driven traffic interpretation in real operational settings.

B. Anomaly Detection

To assess the effectiveness and adaptability of *Revelation*'s anomaly detection stage, we fine-tuned a BERT classifier on the Edge-IIoTset dataset, which encompasses fifteen distinct traffic classes covering reconnaissance, injection, denial-of-service, persistence, and benign activity. The dataset was divided using an 80/20 stratified split, yielding 126,240 samples for training and 31,560 for testing, ensuring balanced representation across all classes. To further evaluate adaptability across heterogeneous IoT environments, the same model was later tested on the TON_IoT dataset [16] (see subsection c), which captures diverse industrial network conditions and attack types.

1) Experimental results

The model achieves 99.88% accuracy, with weighted precision, recall, and F1-score of 99.88%. This reflects consistent performance across both benign and adversarial traffic, demonstrating strong generalization to unseen flows.

a) ROC-AUC Analysis: Figure 4 presents the per-class ROC curves. Every class attains an area under the curve of at least 0.99; in particular, MITM, DDoS over UDP, and DDoS over ICMP achieve exactly 1.00. Normal traffic records approximately 0.9887, and all remaining attack classes fall between 0.9988 and 1.0000. Collectively, these results indicate near-saturated separability across classes and stable discrimination even in the low false-positive operating region.

b) Confusion Matrix: Figure 5 shows consistently high recalls across classes. Perfect recall (100.00%) is achieved for Man-in-the-Middle (MITM), Ransomware, SQL Injection, Distributed Denial-of-Service (DDoS) over UDP, and DDoS over ICMP. Near-perfect results are observed for Uploading, DDoS over HTTP, DDoS over TCP, and Port Scanning (greater than or equal to 99.95% recall). Normal traffic is correctly identified with 99.61% recall, while the lowest recall, 99.50%, occurs for Fingerprinting, reflecting minor confusion among closely related reconnaissance flows. Precision remains uniformly high; the lowest class precision is 99.18% for Ransomware, where recall remains perfect, yielding an F1-score near 99.59%.

To provide a comprehensive view of class-level performance, Table 3 reports the full classification metrics, including precision, recall, and F1 Scores across all fifteen traffic categories. The combination of Figures 4 and 5 with Table 3 demonstrates that the transformer-based anomaly detector achieves highly reliable classification across both benign and adversarial traffic, reinforcing its role as a dependable foundation within the *Revelation* framework.

c) Cross-Dataset Evaluation on TON_IoT: To assess the adaptability of the anomaly detection module across heterogeneous IoT environments, the same BERT classifier was fine-tuned and evaluated on the TON_IoT dataset. TON_IoT is a benchmark dataset reflecting Industry 4.0 environments, containing both benign and malicious traffic with ten diverse attack types, including backdoor, ransomware, XSS, scanning, MITM, and injection. The dataset comprises 211,043 samples spanning backdoor, ransomware, scanning, MITM, injection,

and denial-of-service attacks. We apply the same preprocessing pipeline used for Edge-IIoTset, removing identifier fields such as *src_ip* and *src_port*, resulting in 46 refined features suitable for classification. A stratified split of 80% for training and 20% for testing was used, yielding 42,209 evaluation samples.

classification Report and Confusion Matrix. The model achieved an accuracy of 98.88% with strong macro and weighted averages across all metrics. Table 4 summarizes the class-wise results. High precision and recall are observed for most attack types, with particularly strong performance on ransomware, backdoor, and injection. Fig. 6 shows the confusion matrix, reflecting low misclassification rates. However, the MITM class posed a greater challenge, with an F1-score of 79.84%. Despite a high recall (92.82%), its lower precision (70.04%) indicates frequent false positives, due to limited samples and overlapping traffic patterns with other classes.

ROC AUC Scores. Fig. 7 presents the ROC curves and AUC values for all traffic classes. The model attains near-perfect discrimination across all categories, with AUC values exceeding 0.993. This includes 0.999 for backdoor and 0.9999 for ransomware, underscoring the model’s robustness in identifying complex threats in industrial IoT environments.

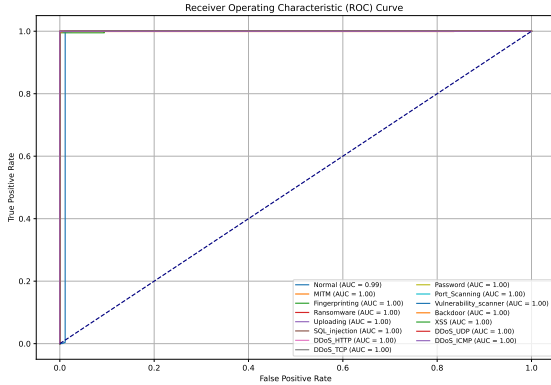


Fig. 4: ROC AUC scores of the fine-tuned BERT model for IoT anomaly detection.

IV. SECURITY AND PRIVACY ANALYSIS

Data Protection and Confidentiality

Revelation safeguards privacy by processing and storing all traffic artifacts locally, including packet captures, logs, flow summaries, and vector embeddings. Only enrichment queries to trusted intelligence sources, such as VirusTotal and Shodan InternetDB, are issued externally. These requests are restricted to confirmed public IP addresses to prevent the disclosure of internal network details. The framework’s reliance on locally hosted open models ensures that sensitive telemetry remains within the administrative perimeter, maintaining full control over data and inference processes.

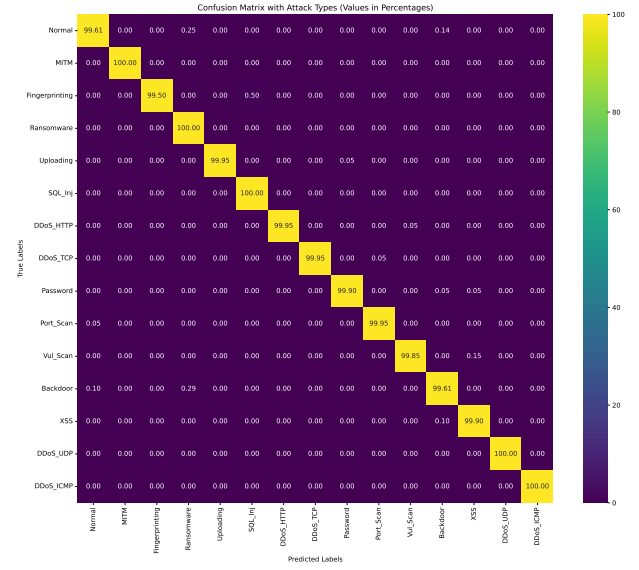


Fig. 5: Confusion matrix showing the classification performance of the fine-tuned BERT model for IoT anomaly detection.

TABLE III: Classification report of the anomaly detector

Class	Precision	Recall	F1-Score	Support
Normal	99.94	99.61	99.77	4,860
MITM	100.00	100.00	100.00	243
Fingerprinting	100.00	99.50	99.75	200
Ransomware	99.18	100.00	99.59	2,185
Uploading	100.00	99.95	99.98	2,054
SQL_Injection	99.95	100.00	99.98	2,062
DDoS_HTTP	100.00	99.95	99.98	2,112
DDoS_TCP	100.00	99.95	99.98	2,050
Password	99.95	99.90	99.92	1,998
Port_Scanning	99.95	99.95	99.95	2,014
Vul_Scanner	99.95	99.85	99.90	2,015
Backdoor	99.51	99.61	99.56	2,039
XSS	99.80	99.90	99.85	2,010
DDoS_UDP	100.00	100.00	100.00	2,900
DDoS_ICMP	100.00	100.00	100.00	2,818
Macro Avg	99.88	99.88	99.88	31,560
Weighted Avg	99.88	99.88	99.88	31,560
Accuracy	99.88%			

Security and model safeguards

The framework enforces strict operational boundaries on both language models and agents to prevent unintended actions or information exposure. The agent’s autonomy is limited to reasoning, retrieval, and controlled tool invocation. It cannot modify configurations, alter network states, or transmit sensitive data. Each operation is logged for accountability, and pre-defined guardrails ensure that prompts and outputs adhere to factual, context-bound, and safety-conscious standards. These mechanisms collectively uphold system integrity, ensuring that intelligent reasoning supports, rather than endangers, secure network operations. Together, these measures make *Revelation* a secure and privacy-preserving framework for interpretable

TABLE IV: Classification report of our fine-tuned BERT on the TON_IoT dataset

Class	Precision (%)	Recall (%)	F1-Score (%)	Support
Normal	99.82	99.76	99.79	10,000
Backdoor	99.85	100.00	99.93	4,000
DDoS	99.65	98.25	98.94	4,000
DoS	99.21	97.70	98.45	4,000
Injection	98.61	95.93	97.25	4,000
MITM	70.04	92.82	79.84	209
Password	99.77	98.75	99.26	4,000
Ransomware	100.00	99.95	99.97	4,000
Scanning	97.98	99.53	98.75	4,000
XSS	95.77	99.08	97.39	4,000
Macro Avg	96.07	98.18	96.96	42,209
Weighted Avg	98.94	98.88	98.90	42,209
Accuracy		98.88%		



Fig. 6: Confusion matrix illustrating the classification performance of our fine-tuned BERT model on the TON_IoT dataset.

IoT traffic analysis.

V. RELATED WORK

Large language models (LLMs) are beginning to reshape how network data is analyzed and explained. Early studies have tested whether general-purpose LLMs can solve networking tasks such as reasoning about topologies, answering diagnostic queries, or interpreting configurations. These works show encouraging results in small or medium networks, but performance degrades as complexity grows, suggesting that purely language-based reasoning is insufficient without structured context [19]. Related efforts treat network traces as text-like inputs, enabling models to narrate packet activity in natural language. While this demonstrates feasibility, the resulting explanations often lack precision because packet data carries limited semantic cues when presented in isolation [20].

To address this gap, researchers have proposed adapting LLMs more directly to traffic data. Approaches such as TrafficLLM introduce traffic-specific tokenization and multi-stage tuning, achieving strong performance in both classification and

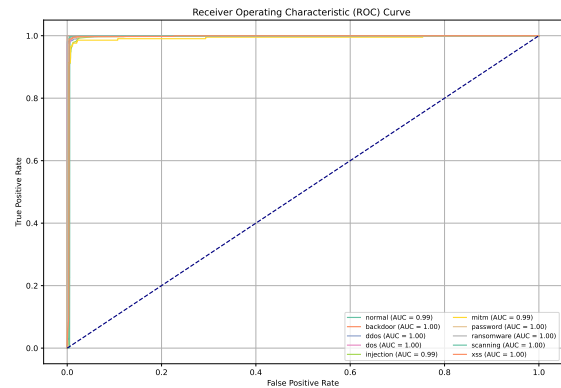


Fig. 7: ROC AUC scores illustrating the performance of our fine-tuned BERT model on the TON_IoT dataset.

generation across diverse scenarios, including unseen traffic types [21]. Agent-based systems extend this idea by using LLMs to coordinate detection pipelines and generate explanations of intrusion events, reporting competitive accuracy even in zero-day settings [22]. Other explainable intrusion detection frameworks employ transformer models to attach rationales to anomaly alerts, improving transparency but raising concerns about cost and efficiency [23]. These efforts advance detection, yet they often reduce network activity to a binary or categorical label and provide only limited insight into benign behaviors that dominate real-world traffic.

Parallel work has explored operational analysis of logs and packet captures. Systems have been developed to automate PCAP failure detection under scarce labels [24] and to assist security analysts with conversational log summarization [25]. More comprehensive designs preprocess and enrich PCAPs, embed them in vector databases, and query them with local LLMs for post-hoc incident reconstruction [24]. Such approaches show that LLM-guided workflows can support forensic tasks, but they typically operate retrospectively and do not unify anomaly detection with a broader interpretation of ongoing network activity. Prior work in IoT administration has also shown that retrieval-augmented generation improves operator answers when paired with anomaly signals, though the focus was on manuals and device documentation rather than raw traffic itself [1].

Beyond traffic and logs, LLMs have demonstrated effectiveness in security intelligence and domain-specific adaptation. In cyber threat intelligence, schema-guided prompting enables accurate extraction of structured events from noisy forum text [26]. At the vulnerability level, models have been tested on generating CVSS scores for CVEs, where they approximate expert scoring but suffer from consistency issues [27]. In protocol analysis, combining retrieval with step-by-step reasoning improves state-machine inference by generating more effective packet seeds for fuzzing [28]. Meanwhile, in telecommunications, both fine-tuned and zero-shot evaluations show that LLMs can capture technical language and intents, although performance is highly variable without specialized training

[29], [30]. These advances confirm that domain adaptation and structured inputs can stabilize LLM outputs; however, their scope is limited to textual corpora or specific protocol logic, rather than the complex mixture of logs, flows, and enriched metadata that characterizes real network environments.

The literature shows that LLMs can reason over network telemetry, adapt to traffic-like representations, detect anomalies, and extract structured knowledge. However, a unified, end-to-end IoT framework remains absent—one that interprets traffic holistically across layers and modalities, integrates device, protocol, flow, and temporal context, incorporates selective threat intelligence, and produces evidence-grounded, operator-actionable explanations for both benign and malicious activity. *Revelation* addresses this gap through a modular processing pipeline with session-aware retrieval and an agent that responds to queries over the indexed corpus.

VI. CONCLUSION

This work presents *Revelation*, an AI-agent-powered framework that transforms raw IoT packet captures into a structured, semantically enriched corpus and enables evidence-grounded question answering. By unifying protocol logs, packet-layer views, flow summaries, a fine-tuned transformer for anomaly detection, and selective threat-intelligence enrichment, *Revelation* supports multi-resolution reasoning that connects routine behavior with malicious activity. An agent mediates interactive analysis by invoking retrieval and web-lookup tools to produce concise and defensible answers to operator queries. Experiments on four representative packet-capture scenarios across six open-source models show that hybrid retrieval improves answer quality and reduces verbosity relative to dense-only retrieval under matched prompts and a shared corpus, while system profiling quantifies runtime and resource overheads consistent with local use. Overall, *Revelation* advances IoT traffic analysis beyond detection by providing evidence-grounded, context-sensitive interpretation suitable for operator-facing analysis.

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